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AEGISEPC Pitch Document

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AEGISEPC

PROJECT INTELLIGENCE & COMPLIANCE PLATFORM

Automating specification RFI resolution and quality compliance auditing for Engineering, Procurement, and Construction (EPC) projects.

Submission: Detailed Project Pitch

Category: ET AI Hackathon 2026

Platform: Google Gemini API

Repository: [Jaswanth258/AI-Data-Centre-EPC-Platform](#)

1. Problem Statement & Context

Infrastructure development, particularly hyperscale data centres, is expanding exponentially (national capacity target growing from 900 MW to 2,700 MW by 2027, representing over \$15 Billion in capital deployment). However, according to McKinsey data, **67% of large EPC projects suffer severe schedule overruns**.

A primary driver of these overruns is **procurement misalignment** and **vendor compliance deviations**. Currently, human quality control engineers spend **3 to 5 days manually cross-referencing** hundreds of parameters in vendor submittals against technical specification documents. Under immense pressure, minor non-conformances — such as an 8-minute battery autonomy rating instead of a 10-minute required minimum — slip through undetected. When these errors are discovered on-site, the resulting equipment rejection and rework create major delays.



Critical Gap: Traditional search and generic chat assistants fail because they do not have exact clause tracing or structured comparison pipelines, leading to hallucinations and inaccurate compliance audits.

2. The AegisEPC Solution

AegisEPC is an AI-powered intelligence platform that operates directly on specification structures without complex vector databases or chunking. It offers two core agent-driven interfaces:



RFI & Knowledge Copilot

Allows engineers to query project specifications in natural language. The agent returns a cited answer alongside interactive citation links that open the exact clause text in a side-by-side viewer. Includes per-document chat memory isolation.

Quality Compliance Agent

Automates vendor submittal reviews. Uploading a PDF submittal extracts all technical parameters, matches them dynamically to the correct spec clause, computes compliance status (Compliant/Deviation), highlights warning/severity levels, and exports CSV logs.

3. Technical Innovation: Clause-Tree Retrieval

Standard RAG systems chunk documents into arbitrary 500-token blocks, which often breaks up clauses and loses hierarchical context. AegisEPC addresses this with **Clause-Tree Retrieval**:

- 1

Deterministic Ingestion: The parser structures specifications into a nested JSON structure preserving headers, subheadings, and lists (e.g., Section 3.1 steel requirements are explicitly structured as subnodes of Section 3).
- 2

2-Step Navigation Loop: To answer a query or check compliance, the LLM first analyzes the Table of Contents (TOC) to identify the relevant clause header. The backend then fetches the exact text of that clause directly from the structured tree.
- 3

Verifiable Traceability: Because retrieval is based on clause structures rather than vector similarity, every answer is guaranteed to reference the correct document section without hallucinated source citations.

4. Live Audit Results & Case Study

When running the compliance agent on a standard UPS Submittal against project specifications, AegisEPC successfully processes the audit in under **8 seconds**, yielding the following parameters:

PARAMETER	SUBMITTED VALUE	SPECIFICATION REQUIREMENT	STATUS	SEVERITY
Battery Autonomy	8 minutes	Min 10 minutes at full load	DEVIATION	Critical
Short-circuit Rating	20 kA	Min 25 kA for 3 cycles	DEVIATION	Critical
Nominal Voltage	415 V	415 V, 3-phase, 4-wire	COMPLIANT	None



5. Business Impact & Scalability

99.8%

Reduction in Submittal Audit Time

100%

Explainable Audit Trail Traceability

Zero

Vector Database Infrastructure Costs

By accelerating standard engineering review cycles from days to seconds, AegisEPC removes bottleneck delays from procurement pipelines, preventing project schedule overruns and ensuring zero cost impact from post-delivery equipment rejections.